

United States GDP Regimes, Fiscal Structure, and Distributional Growth

A Reproducible Data-Science Report, 1920–Present

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`us-gdp-regime-1920`

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Abstract

This report summarizes a reproducible empirical workflow for long-run United States real GDP growth from 1920 onward. The analysis combines a Maddison-based real GDP proxy, FRED/BEA validation data, piecewise growth-regime detection, federal fiscal context, tax-regime event dynamics, and distributional measures linking GDP per capita to real worker earnings. The headline statistical result is that a single long-run trend fits the log GDP level well, but annual growth is better read as a sequence of regimes. The distributional result is equally important: aggregate real GDP per capita rose much faster than real median weekly earnings after the late 1970s, so GDP growth alone is an incomplete proxy for broad buying power. Fiscal and tax variables add useful context, but the report distinguishes descriptive timing, lagged association, and causal identification throughout.

Keywords: United States GDP, growth regimes, public debt, tax policy, real wages, distributional analysis, local projections, reproducible research

All generated tables and figures are reproducible from the Python pipeline in this repository. Raw downloaded data are intentionally excluded from version control.

1 Executive Summary

The project began with a simple question: whether the United States grew at one roughly constant speed after 1920, or whether the data support distinct growth regimes. The answer is mixed in the way macroeconomic time series usually are: the log level of real GDP has a strong long-run trend, but the annual growth process is segmented by depression, war, demobilisation, and a long postwar period.

The fitted log-trend model implies an annualized trend growth rate of approximately 3.16% and an R^2 of 0.987. The piecewise growth model selects five broad regimes in the Maddison-derived annual GDP series: 1921–1928, 1929–1933, 1934–1944, 1945–1949, and 1950–2022. These regimes are descriptive statistical summaries; they are not automatic causal explanations.

The expanded fiscal and distributional layers change the interpretation. Federal debt, deficits, and tax laws help structure the historical narrative, but they are endogenous to the same shocks

that affect GDP. More importantly, GDP per capita and real worker earnings diverged strongly after 1979. Indexed to 1979, real GDP per capita was about 208.6 in 2023, while real median weekly earnings were about 110.4. In other words, aggregate output per person more than doubled, while the median weekly earnings index increased by only about 10.4% over the same base.

2 Data and Reproducibility

2.1 Core GDP Series

The primary GDP series is built from the Maddison Project Database 2023 by multiplying United States real GDP per capita by population. This produces a long-run real GDP proxy suitable for growth-rate and log-trend analysis from 1920 onward. The modern validation source is FRED/BEA annual real GDP (GDPCA), available from 1929 onward.

The source-validation table compares Maddison-derived growth with FRED/BEA growth from 1930 to 2022. The overlap has 93 annual observations, a growth-rate correlation of 0.955, a mean absolute difference of 0.626 percentage points, and an RMSE of 1.471 percentage points. This is close enough for regime-level analysis, while still reminding the reader that Maddison historical estimates and BEA national accounts are not the same object.

2.2 Fiscal, Tax, and Distributional Data

The fiscal context layer uses FRED/OMB federal debt, receipts, outlays, deficit, and interest-outlay ratios as percent of GDP. The tax-regime event layer uses a curated catalog of broad federal tax-law changes, with signed ordinal shock values and narrative classifications. The distributional layer adds BEA real GDP per capita, BLS real median weekly earnings, BLS real hourly compensation, BEA federal receipt composition, IRS top and bottom statutory income-tax rates, and BLS Consumer Expenditure federal income-tax rates by income quintile.

These series answer different questions. GDP per capita measures aggregate real output per person. Median weekly earnings measure the real earnings of the median full-time wage and salary worker. Federal income-tax rates by Consumer Expenditure income quintile measure average federal income-tax burdens for consumer units, not the full incidence of the tax-and-transfer system.

3 Methods

3.1 Trend Model

The long-run trend is estimated on log real GDP:

$$\log(GDP_t) = \alpha + \beta \cdot year_t + \epsilon_t. \quad (1)$$

The annualized trend growth rate is computed as $\exp(\beta) - 1$. This captures average exponential growth, but it intentionally does not model nonlinear business-cycle dynamics.

3.2 Piecewise Growth Regimes

Growth regimes are estimated on annual percent changes:

$$growth_t = \mu_j + u_t, \quad t \in \text{regime } j. \quad (2)$$

The algorithm searches over candidate breakpoints with dynamic programming and selects the segmentation with a BIC-style penalty. A segment is labelled “above mean” if its mean growth rate exceeds the full-sample mean. This is a descriptive label, not a welfare or causality statement.

3.3 Dynamic Tax Effects

The dynamic tax-effects layer uses two complementary designs. First, local projections estimate the association between a signed tax shock and future GDP growth at each horizon:

$$growth_{t+h} = \alpha_h + \beta_h shock_t + \gamma_h growth_{t-1} + \gamma'_h growth_{t-2} + e_{t+h}. \quad (3)$$

Second, distributed-lag regressions include current and lagged tax shocks in a single annual growth equation. Both designs estimate delayed dynamics, but they do not solve identification by themselves. The report therefore separates all catalogued tax events from a smaller subset classified as plausibly exogenous long-run reforms.

3.4 Distributional Analysis

The distributional layer indexes real GDP per capita, real median weekly earnings, and real hourly compensation to a common base year:

$$index_{x,t} = 100 \times \frac{x_t}{x_{base}}. \quad (4)$$

The main gap measure is:

$$gap_t = index_{GDPpc,t} - index_{medianearnings,t}. \quad (5)$$

Tax-burden shift is approximated with three sets of proxies: receipt composition, statutory top-bottom income-tax rate spread, and Consumer Expenditure federal income-tax rates by income quintile. These proxies do not constitute a full incidence model, but they directly address whether the tax structure moved away from more progressive bases and whether this aligns with GDP growth or worker earnings.

4 Results

4.1 Long-Run GDP Trend and Growth Regimes

Figure 1 shows the log real GDP path and fitted linear trend. The model fit is high, but the residuals and annual growth series show why the trend alone is not an adequate historical summary.

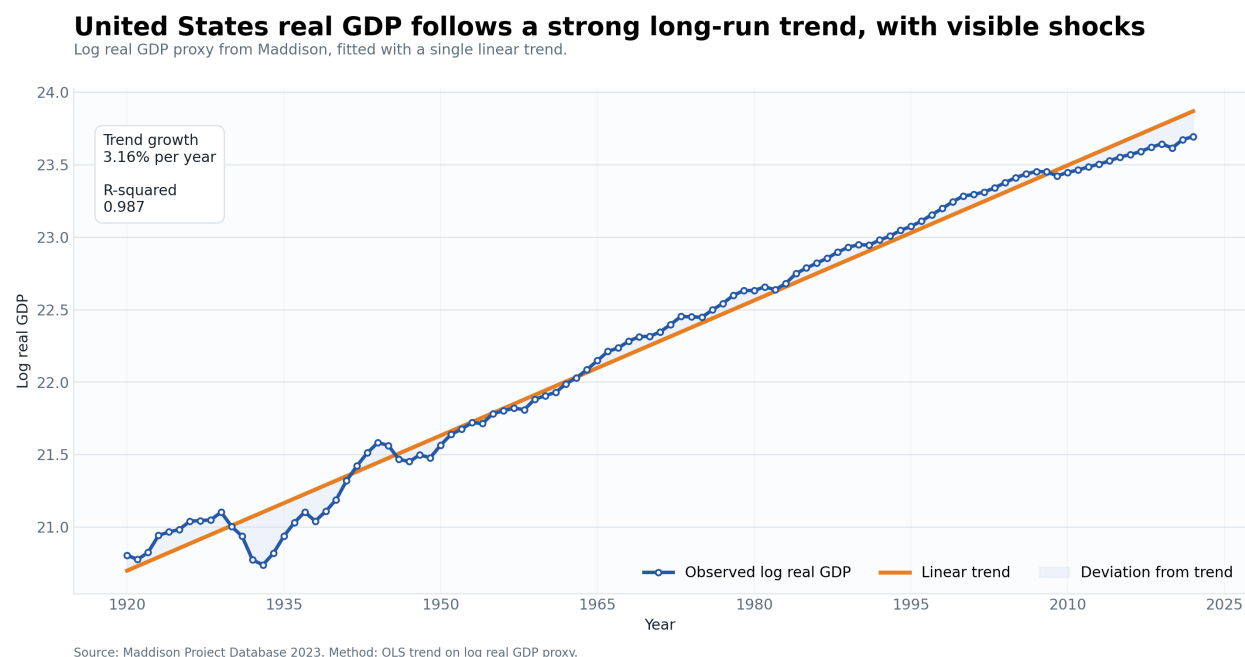


Figure 1. Log real GDP trend from the Maddison-derived United States real GDP proxy. The fitted trend summarizes average exponential growth, while deviations identify periods that require historical interpretation.

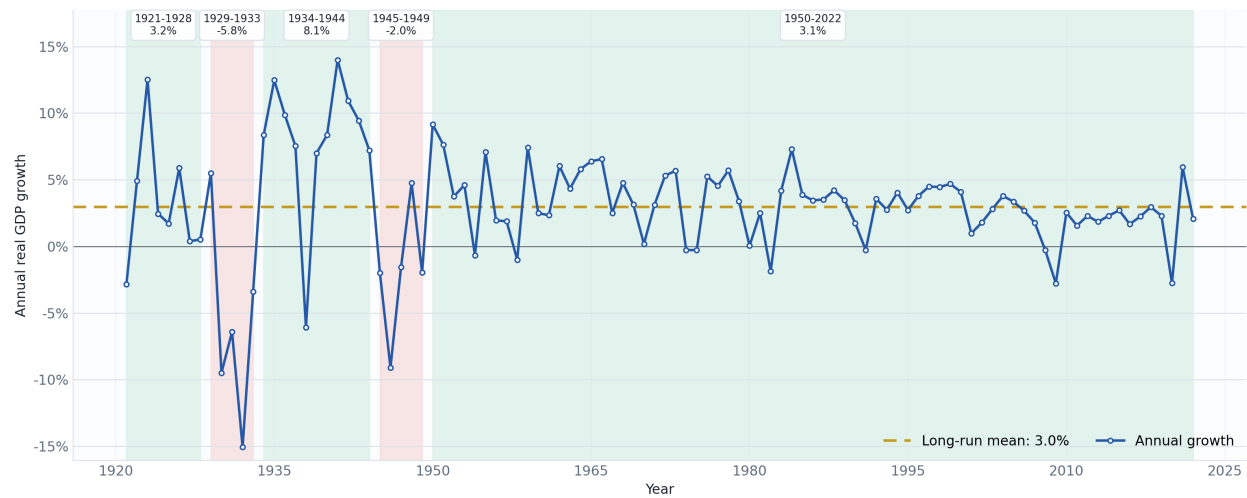
Table 1 summarizes the fitted growth regimes. The severe 1929–1933 contraction and the 1934–1944 rebound/mobilisation period dominate the early sample. The post-1950 period is long and relatively stable by this model’s penalty criterion, with mean annual growth of approximately 3.11%.

Table 1. Fitted annual GDP growth regimes

Start	End	Years	Mean growth	Label
1921	1928	8	3.19%	Above mean
1929	1933	5	-5.78%	Below mean
1934	1944	11	8.10%	Above mean
1945	1949	5	-1.98%	Below mean
1950	2022	73	3.11%	Above mean

Growth regimes separate temporary volatility from sustained phases

Piecewise mean growth segments selected on annual real GDP growth.



Source: Maddison Project Database 2023. Green/red bands indicate above/below full-sample mean segment growth.

Figure 2. Piecewise annual real GDP growth regimes. Bands are statistical segments, not claims that one historical event caused the entire segment.

4.2 Fiscal Context

Adding federal fiscal ratios changes the narrative from a pure growth story to a macro-fiscal story. Debt and deficit measures vary sharply across wars, recessions, and crisis periods. Same-year correlations between fiscal ratios and GDP growth are modest. In the generated panel, gross debt-to-GDP has a same-year correlation with GDP growth of about -0.317 and a one-year-lag correlation of about -0.261 . Receipts, outlays, and deficits have smaller simple correlations with annual growth.

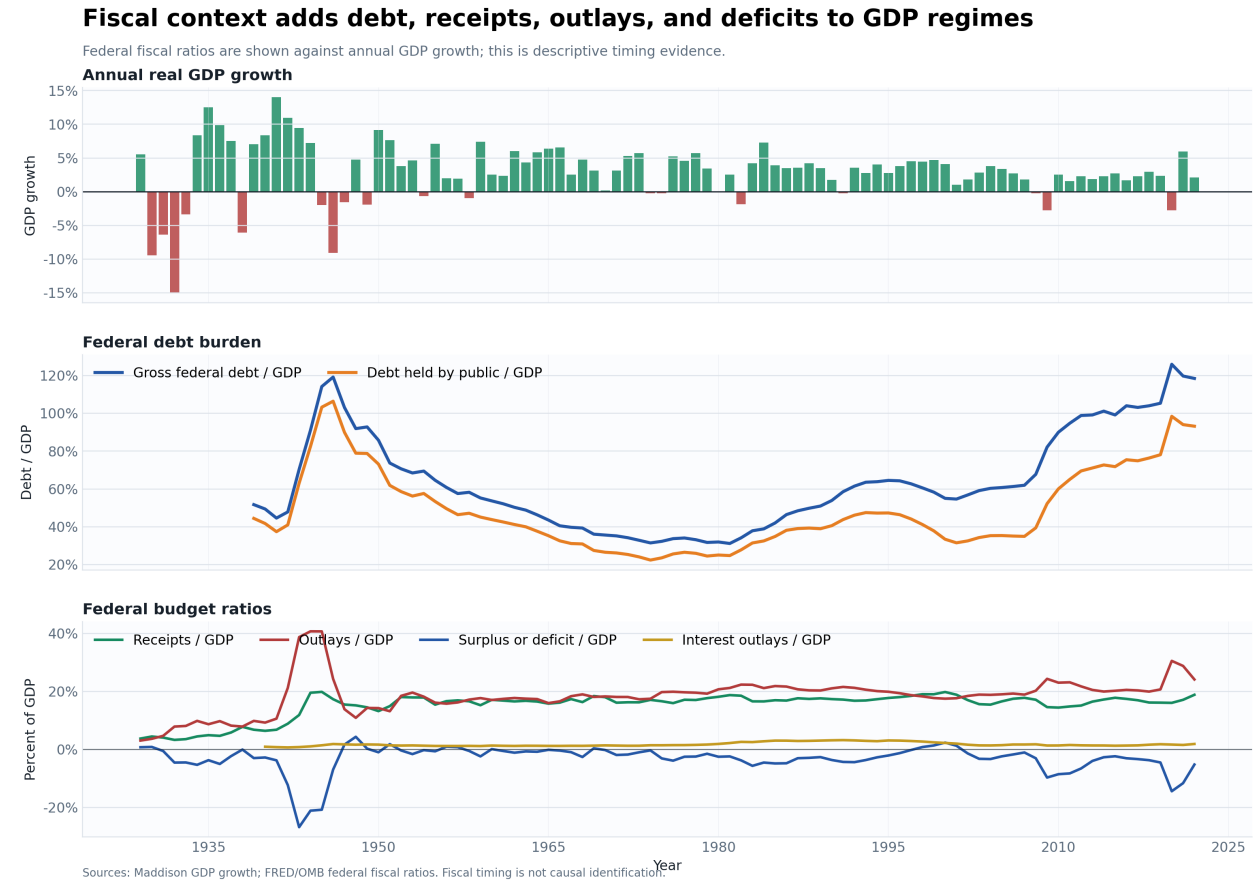


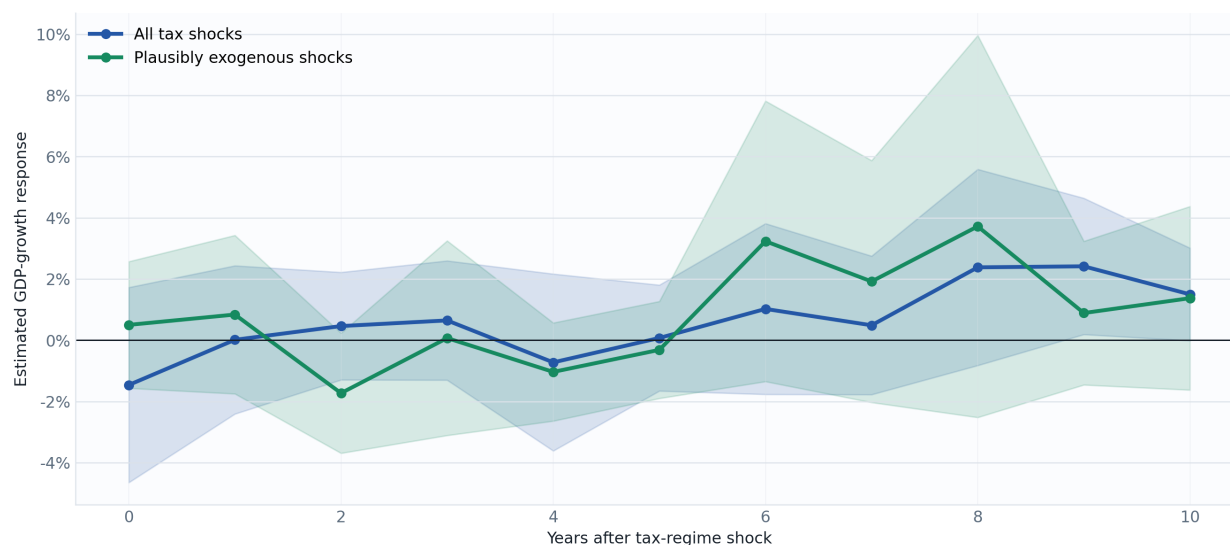
Figure 3. Federal debt, budget ratios, and annual GDP growth. The figure is a timing and context device; it does not identify fiscal causality.

4.3 Tax-Regime Dynamics

The local-projection estimates in Figure 4 show delayed paths after signed tax-regime shocks. Confidence intervals are wide, especially for the plausibly exogenous subset, because there are few major long-run reform events in the historical sample. The result should therefore be read as a dynamic screening exercise rather than a definitive tax multiplier estimate.

Tax-shock local projections show delayed GDP-growth associations

Lines estimate future annual GDP growth at each horizon after a signed tax-regime shock.



Source: Maddison GDP growth and curated tax-shock catalog. Causal interpretation is limited to plausibly exogenous shocks and remains model-dependent.

Figure 4. Local-projection estimates of GDP-growth associations after signed tax-regime shocks. The green line restricts the treatment series to events classified as plausibly exogenous long-run reforms.

The most important methodological improvement over a same-year event study is that the model allows effects to appear with delay. The data do not support a simple one-year story. Some horizons are negative, some positive, and most intervals overlap zero. That pattern is consistent with the fact that tax laws include anticipation, phase-ins, temporary provisions, and heterogeneous channels across households and firms.

5 Tax Burden Shift and Real Wages

5.1 GDP Per Capita versus Worker Buying Power

Figure 5 is central to the distributional interpretation. Using 1979 as the common index base, real GDP per capita reached about 208.6 by 2023, while real median weekly earnings reached about 110.4. Real hourly compensation rose more than median weekly earnings, but still much less than GDP per capita. This means aggregate output per person can increase while the median worker's measured real earnings grow slowly.

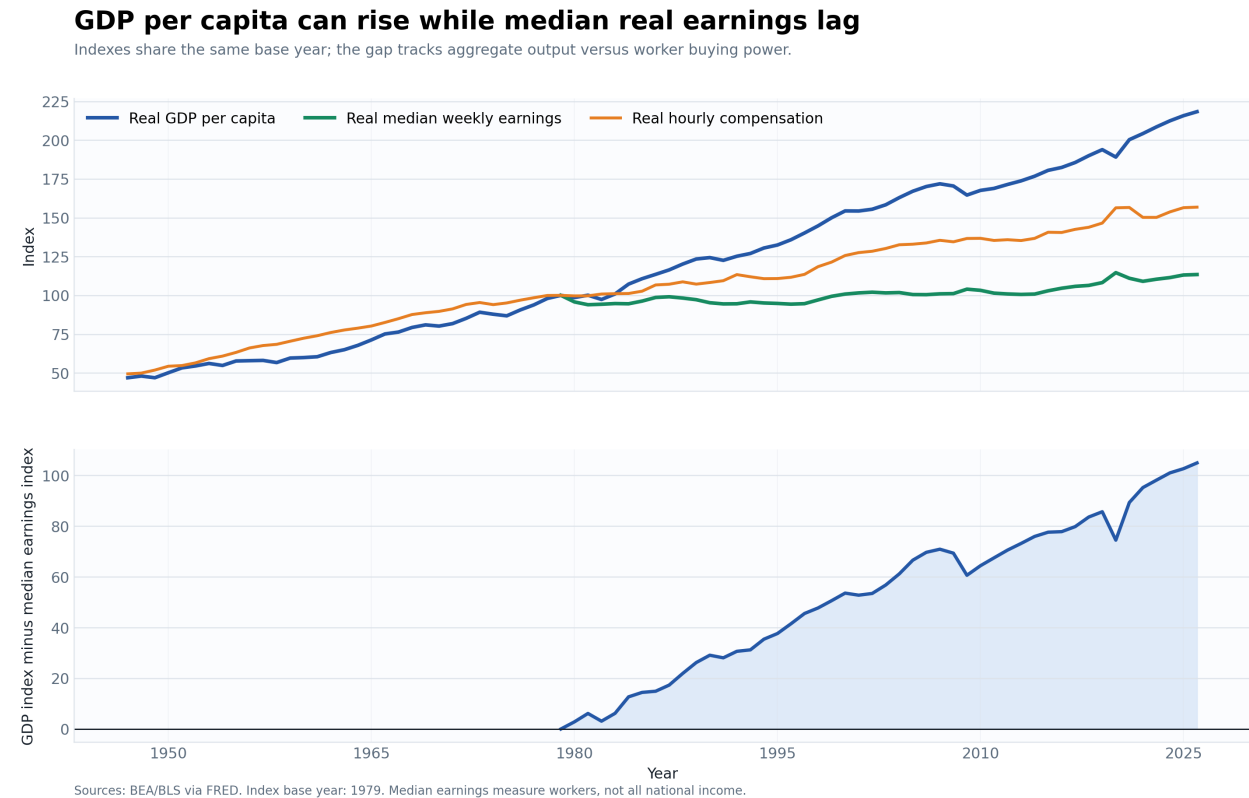


Figure 5. Real GDP per capita, real median weekly earnings, and real hourly compensation indexed to 1979. The widening gap shows why GDP growth is not equivalent to broad worker buying-power growth.

This does not mean most people necessarily became poorer in every year. It means the aggregate growth measure and the median wage-worker measure diverged substantially. A welfare analysis would also need hours worked, household composition, transfers, taxes, fringe benefits, asset income, housing costs, and consumption prices by income group.

5.2 Tax Burden Shift Proxies

Figure 6 combines three views of tax burden. The first panel shows the composition of selected federal receipts. Social-insurance contributions rose from about 15.1% of the selected federal receipt base in 1947 to about 35.4% in 2023. Corporate income taxes fell from about 29.6% to about 8.4% over the same endpoint comparison. This is consistent with a long-run movement away from corporate income taxes and toward social-insurance financing.

Tax burden shift proxies combine receipt composition and quintile tax rates

Social-insurance receipts proxy less-progressive payroll taxation; quintile rates show income-tax burden.

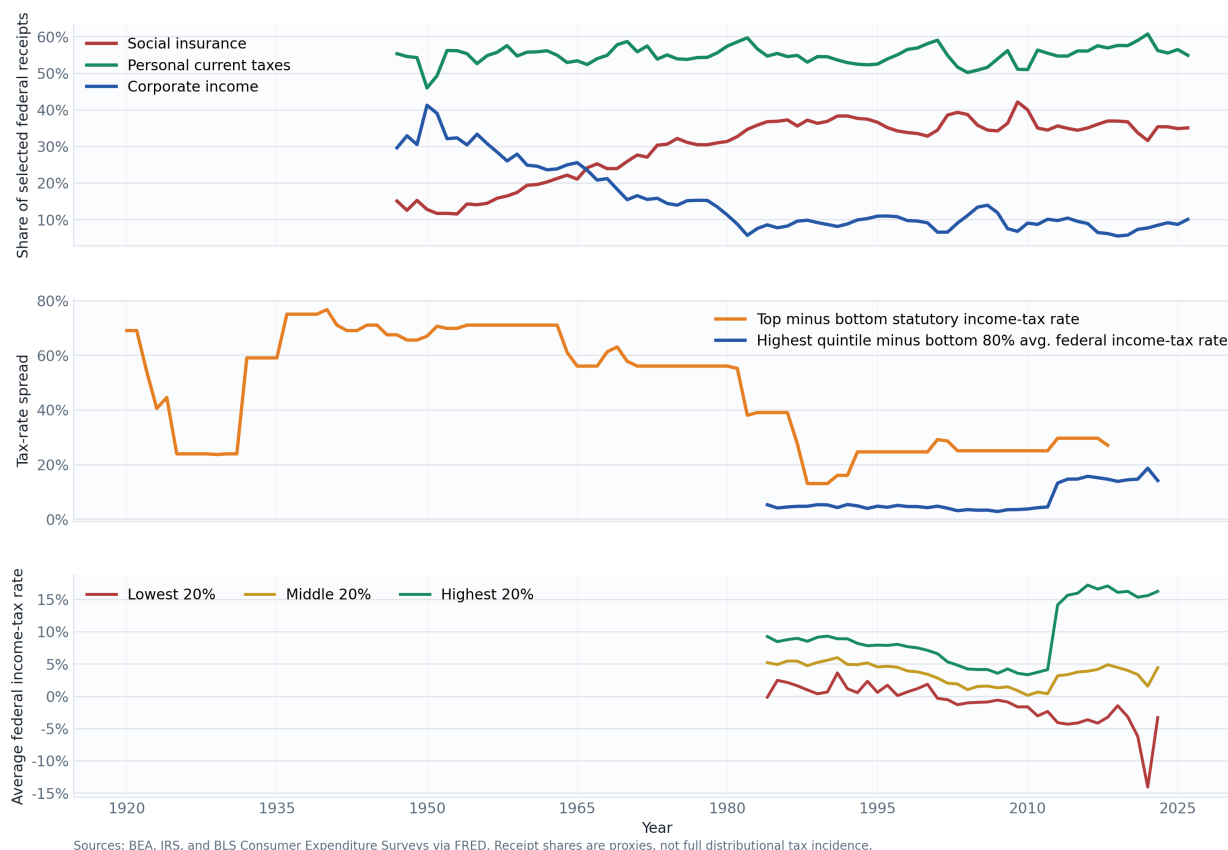


Figure 6. Tax burden shift proxies. Receipt composition captures the rise of social-insurance contributions relative to corporate income taxes; quintile rates show average federal income-tax rates by income group where Consumer Expenditure data are available.

The Consumer Expenditure quintile panel begins in 1984. In that year, the highest income quintile's federal income-tax rate exceeded the average of the bottom four quintiles by about 5.25 percentage points. By 2023, that spread was about 14.10 percentage points. This particular income-tax measure therefore does not show a simple shift of federal income-tax burden away from the top quintile. The broader burden story is more nuanced: payroll-style financing became more important, corporate income-tax receipts became less important, and income-tax progressivity by quintile remained substantial.

5.3 Association Screen

The lagged association screen regresses GDP growth, real median weekly earnings growth, and the GDP/wage gap on one-year-lagged burden-shift proxies. These are single-predictor models with HAC standard errors. The strongest pattern in the current output is not a robust GDP-growth claim; it is that the GDP/wage gap is statistically associated with some tax progressivity proxies. For example, the one-year-lag statutory top-bottom rate spread has a negative coefficient in the GDP/wage-gap regression, while the quintile rate spread has a positive coefficient. These associations should be interpreted cautiously because the models are low-dimensional and do not control for

technology, unionization, trade exposure, monetary policy, inflation, demographic composition, or sectoral change.

6 Limitations

The main limitation is identification. Breakpoints, fiscal ratios, tax laws, and wages all move with macroeconomic conditions. Recessions change tax receipts and policy decisions; wars change debt, output, and tax rates; inflation and monetary policy affect both GDP and real earnings. The tax-shock catalog partly addresses this by separating plausibly exogenous reforms from deficit, wartime, and countercyclical changes, but the classification remains historical and sample sizes are small.

The second limitation is measurement. Maddison is useful for long-run history but is not identical to BEA national accounts. Median weekly earnings are not the same as household income or total compensation. Consumer Expenditure quintile tax rates cover federal income-tax amounts, not the full tax-and-transfer system. Receipt composition is a proxy for burden shifting, not a complete incidence analysis.

The third limitation is temporal coverage. Some modern FRED series include annualized or partial latest-year observations, while Consumer Expenditure quintile data currently end earlier. The report therefore reports explicit coverage by series and avoids implying that every dataset covers the same complete historical window.

7 Conclusion

The GDP regime analysis shows that the United States did not grow at one constant annual speed after 1920. A log trend summarizes the long-run level well, but annual growth is better read through regimes. Fiscal and tax data enrich the narrative, especially by showing how debt, deficits, tax-law changes, and federal receipt composition line up with macroeconomic periods.

The distributional extension changes the substantive interpretation. Real GDP per capita and real median weekly earnings diverged strongly after 1979. That does not negate GDP growth; it clarifies who may have participated in it. From a data-science perspective, the correct conclusion is not that GDP is useless, but that GDP is incomplete. A credible assessment of economic regimes should report aggregate output, fiscal structure, tax burden proxies, and worker purchasing power together.

Reproducibility Statement

The report is generated from the repository pipeline:

- `us-gdp-regimes run --config config/default.yaml`
- `us-gdp-regimes make-fiscal-context --config config/default.yaml`
- `us-gdp-regimes make-tax-effects --config config/default.yaml`
- `us-gdp-regimes make-distributional-analysis --config config/default.yaml`

The relevant notebooks are 01 through 06 in the `notebooks/` directory. Generated raw data, model CSVs, figures, and article assets are reproducible outputs and are not committed as raw source data.

References

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